

Scalability & Future Plan

Date	3 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie – AI Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently runs locally using Uvicorn and is not deployed for large-scale or public access.	Limits the number of users and prevents centralized access to the application.	High
2	The current version does not use a database and does not save user history or learning progress between sessions.	Prevents persistent learning records, progress tracking and data-driven personalization.	High
3	Response speed and availability depend on the Gemini API, including API rate limits and temporary service errors. Performance measurements have not yet been executed.	High request loads may cause delays, retries or temporary failures, and scalability is dependent on the external AI service.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	The application currently runs locally with Uvicorn and has no horizontal scaling or load balancing.	Deploy EduGenie to a suitable cloud/server environment and use multiple application instances with load balancing to support a larger number of concurrent users.
2	Data Storage	No database is used in the current version, and user history or learning progress is not stored.	Introduce a persistent database to store user profiles, learning history, quiz results and progress data. This will support dashboards and future personalized learning features.
3	Performance	Response time depends on Gemini API behaviour. The application uses a 45-second timeout and retry handling for temporary errors. No measured performance results are currently available.	Use asynchronous processing, response caching where appropriate, API rate limiting, performance monitoring and suitable model/service optimization. Load testing should also be executed before large-scale deployment.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	The Gemini API key is stored through an environment variable, but the current local prototype does not provide user authentication or a production security layer.	For future deployment, add user authentication and authorization, secure secret management, HTTPS, access controls and appropriate protection for stored user data and API credentials.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deployment beyond local execution, persistent database, user authentication and basic security improvements.	Initial scale-up phase	Makes EduGenie suitable for broader access while providing secure and persistent user data management.
Phase 2	Progress tracking dashboards, gamification with badges and learning streaks, and adaptive learning paths based on learner performance.	Short-term enhancement phase	Enables learners to monitor their learning journey and receive more personalized learning guidance.
Phase 3	Voice interaction, multilingual learning support, and a mobile application with offline capability.	Medium-term enhancement phase	Improves accessibility, language coverage and learning availability across devices and connectivity conditions.
Phase 4	Group study sessions, teacher/parent dashboards, LMS integration, image/PDF input and smart notifications.	Long-term enhancement phase	Expands EduGenie into a more collaborative, integrated and feature-rich learning platform.

Note: The above roadmap represents future enhancements documented in the project report. These features are not implemented in the current version. No specific calendar dates for these future phases are defined in the Final Report, so phase-based timelines are used instead of inventing dates.